# You Did Not Just Read This

or did you?

# Quantum Computing

## Lecture 3: Teleportation, Superdense Coding, Quantum Algortihms


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**Department of Computer Science & Engineering**
**University of Washington**





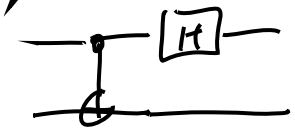

# Summary of Last Lecture

qubit




$|\psi\rangle$

2 qubits



# Final Examination

# Quantum Teleportation

Alice wants to send her qubit to Bob.
She does not know the wave function of her qubit.

Alic
e



Bob



$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Can Alice send her qubit to Bob using classical bits?

Since she doesn't know $|\psi\rangle$ and measurements on her state do not reveal $|\psi\rangle$, this task appears impossible.

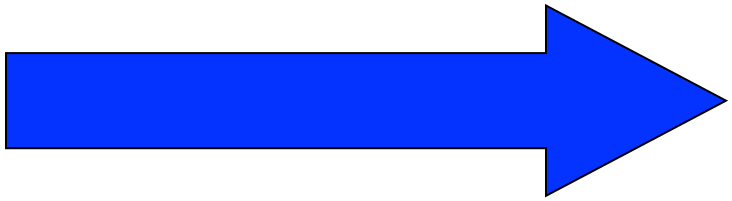
# Quantum Teleportation

Alice wants to send her qubit to Bob.
She does not know the wave function of her qubit.

Alic
e



classical communication



Bob



$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Suppose these bits contain information about $|\psi\rangle$

Then Bob would have information about $|\psi\rangle$ as well as the qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

This would be a procedure for extracting information from $|\psi\rangle$ without effecting the state

# ~~Quantum~~ Teleportation Classical

Alice wants to send her probabilistic bit to Bob using classical communication.

Alic
e



Bob



$$v = \begin{bmatrix} p \\ 1-p \end{bmatrix}$$

She does not wish to reveal any information about this bit.

# Classical Teleportation

(a.k.a. one time pad)

Alic
e



$$v = \begin{bmatrix} p \\ 1-p \end{bmatrix} \begin{matrix} 0 \\ 1 \end{matrix}$$

$$e = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{1}{2} \end{bmatrix}$$

50 %
00
50 % 11

Bob



Alice and Bob have two perfectly correlated bits

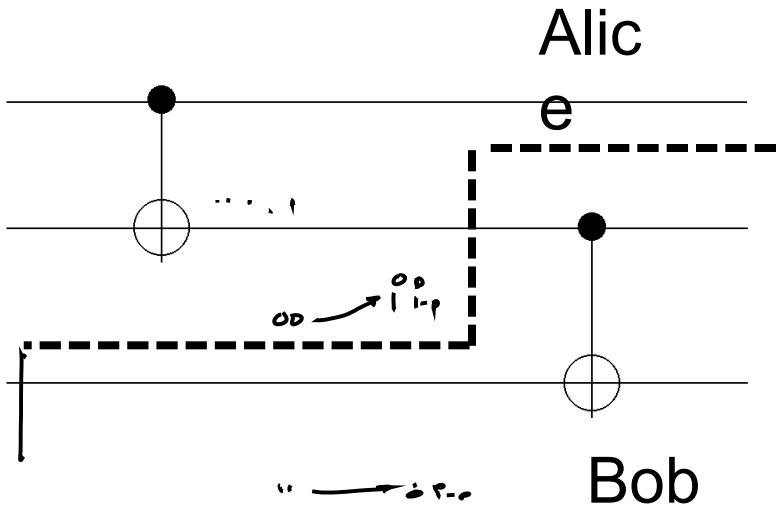
Alice XORs her bit $v$ with the correlated bit and sends the result to Bob.

Bob XORs his correlated bit with the bit Alice sent and thereby obtains a bit with probability vector $v$ .

# Classical Teleportation Circuit

$$v = \begin{bmatrix} p \\ 1-p \end{bmatrix}$$

$$e = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{1}{2} \end{bmatrix}$$




$$v = \begin{bmatrix} p \\ 1-p \end{bmatrix}$$

| | | | |
|---|---|---|---|
| 000 | $\frac{p}{2}$ | $\frac{p}{2}$ | $\frac{p}{2}$ |
| 001 | 0 | 0 | 0 |
| 010 | 0 | 0 | $\frac{p}{2}$ |
| 011 | $\frac{p}{2}$ | $\frac{p}{2}$ | 0 |
| 100 | $\frac{1-p}{2}$ | 0 | 0 |
| 101 | 0 | $\frac{1-p}{2}$ | $\frac{1-p}{2}$ |
| 110 | 0 | $\frac{1-p}{2}$ | 0 |
| 111 | $\frac{1-p}{2}$ | 0 | $\frac{1-p}{2}$ |

No information in transmitted bit:

$$\begin{array}{ll} 000 & \frac{p}{2} \\ 001 & 0 \\ 010 & 0 \\ 011 & \frac{p}{2} \\ 100 & 0 \\ 101 & \frac{1-p}{2} \\ 110 & \frac{1-p}{2} \\ 111 & 0 \end{array}$$

transmitted bit

$$\begin{bmatrix} \frac{p}{2}+0+0+\frac{1-p}{2} \\ 0+\frac{p}{2}+\frac{1-p}{2}+0 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

And it works:

$$\begin{array}{ll} 000 & \frac{p}{2} \\ 001 & 0 \\ 010 & \frac{p}{2} \\ 011 & 0 \\ 100 & 0 \\ 101 & \frac{1-p}{2} \\ 110 & 0 \\ 111 & \frac{1-p}{2} \end{array}$$

Bob's bit

$$\begin{bmatrix} \frac{p}{2}+\frac{p}{2}+0+0 \\ 0+0+\frac{1-p}{2}+\frac{1-p}{2} \end{bmatrix} = \begin{bmatrix} p \\ 1-p \end{bmatrix}$$

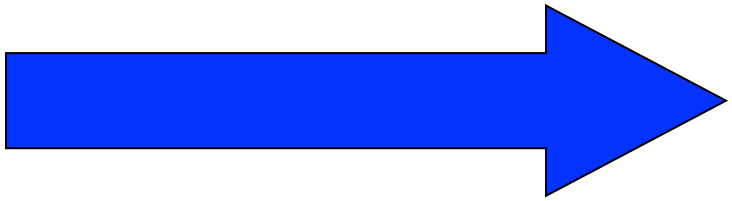
# Quantum Teleportation

Alice wants to send her qubit to Bob.
She does not know the wave function of her qubit.

Alic
e



classical communication



Bob



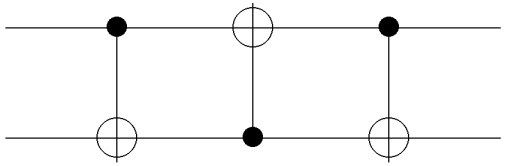
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

allow them to share the entangled state:

$$|e\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi_+\rangle$$

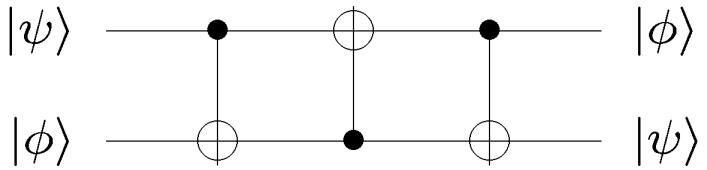
# Deriving Quantum Teleportation

Our path: We are going to “derive” teleportation



$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

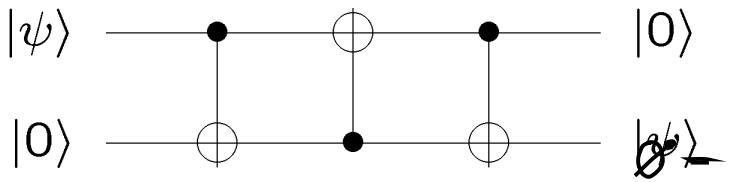
“SWAP”

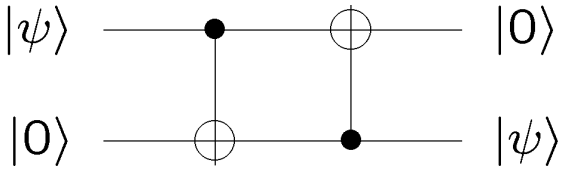



“Alice”

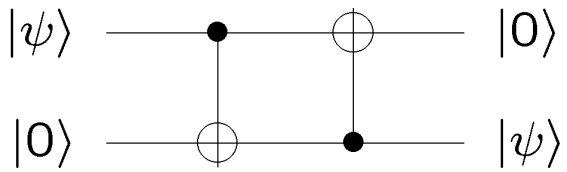
“Bob”

Only concerned with from Alice to Bob transfer

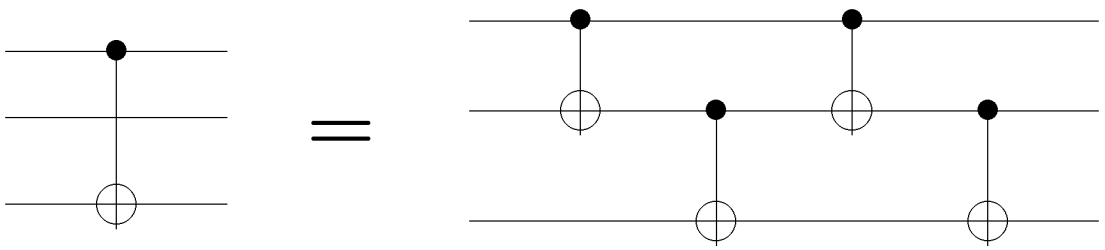


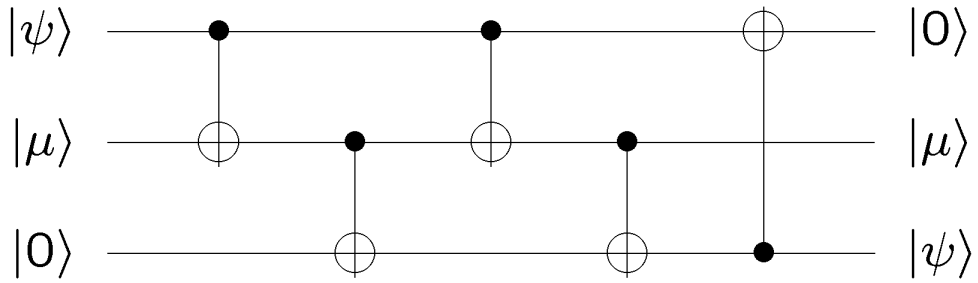
# Deriving Quantum Teleportation




Need some way to get entangled states

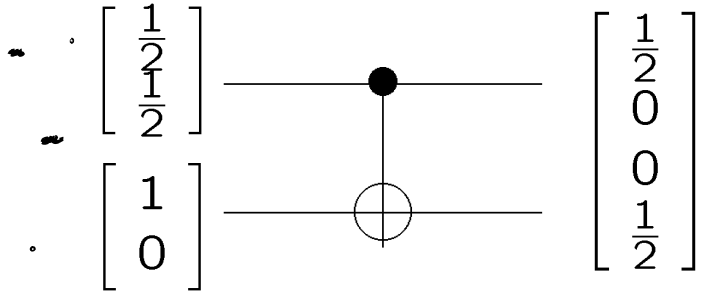


new equivalent circuit:

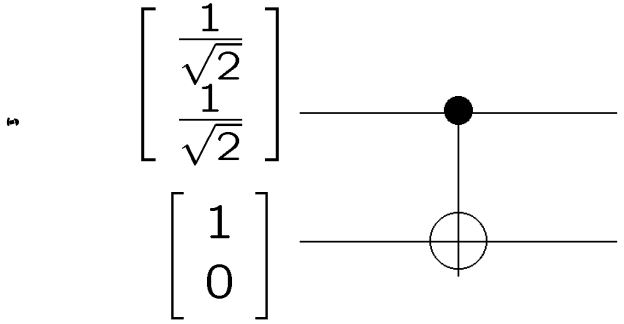


# Deriving Quantum Teleportation

How to generate classical correlated bits:



Inspires: how to generate an entangled state:



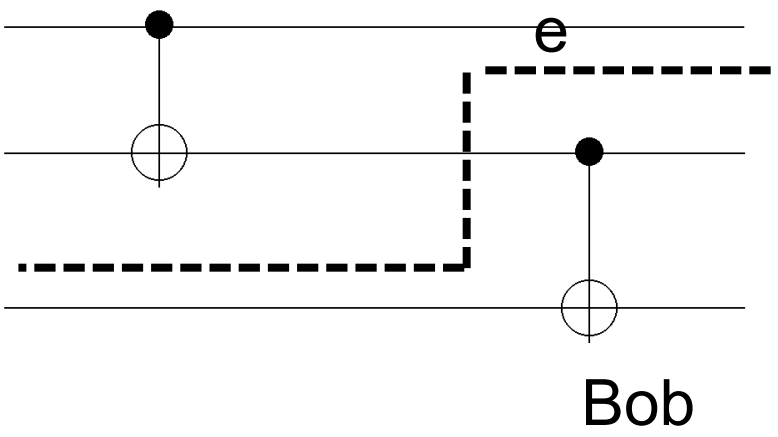
$$= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

# Deriving Quantum Teleportation

Classical Teleportation

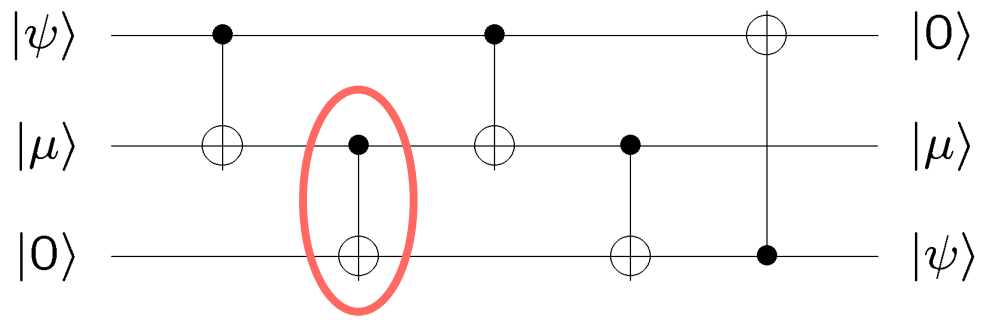
$$v = \begin{bmatrix} p \\ 1-p \end{bmatrix}$$

$$e = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{1}{2} \end{bmatrix}$$



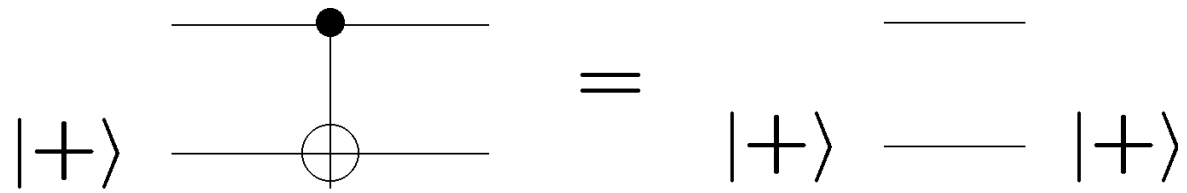

$$r = \begin{bmatrix} \frac{p}{2} \\ \frac{p}{2} \\ \frac{1-p}{2} \\ \frac{1-p}{2} \end{bmatrix}$$

$$v = \begin{bmatrix} p \\ 1-p \end{bmatrix}$$

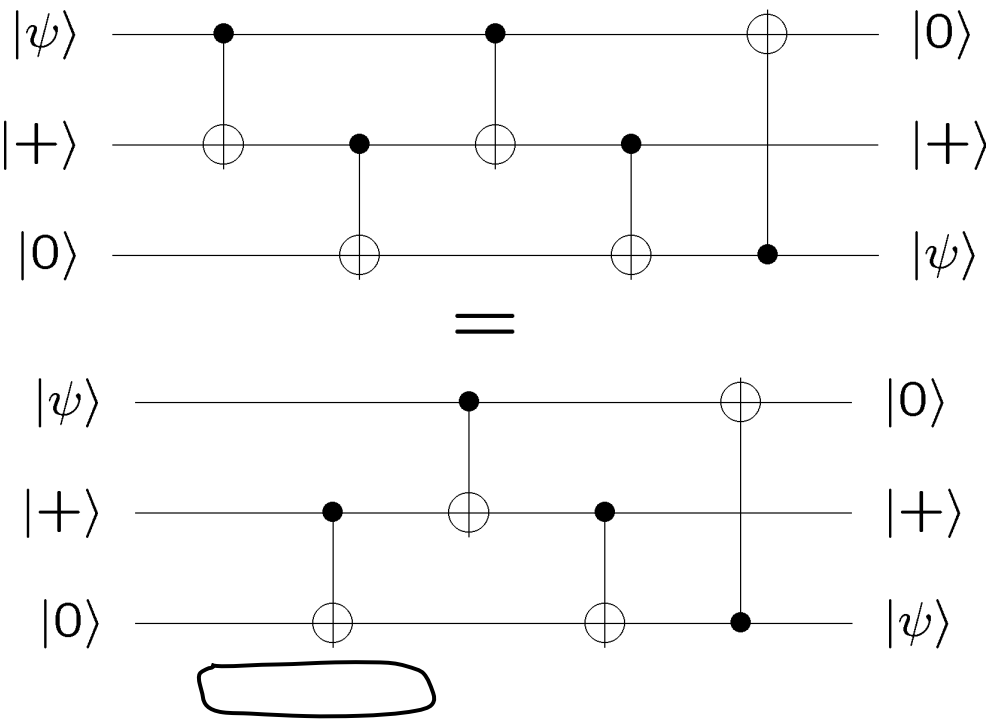



like to use generate entanglement

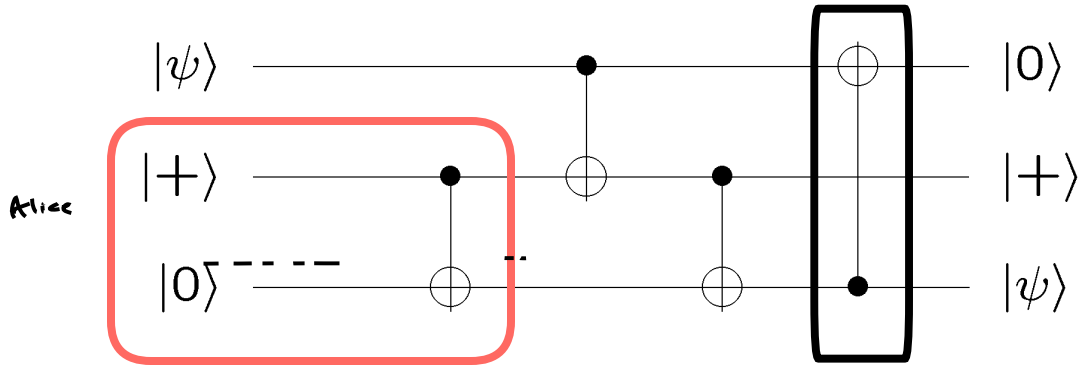
# Deriving Quantum Teleportation




$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \qquad X|+\rangle = |+\rangle$$



# Deriving Quantum Teleportation



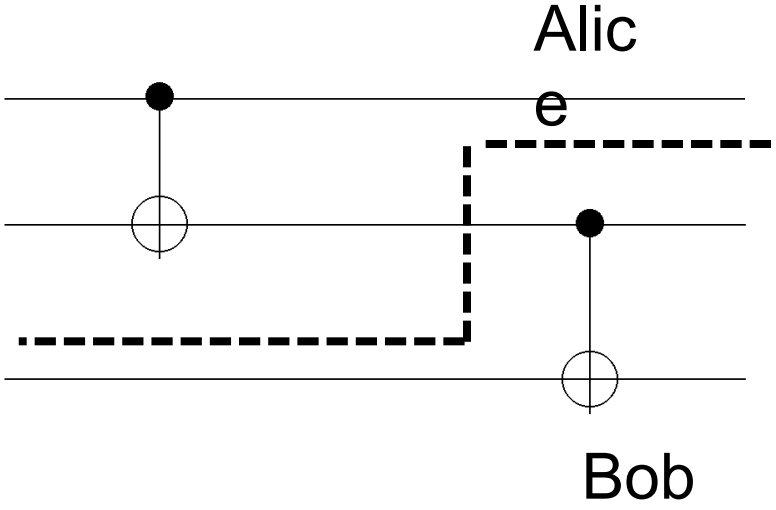

$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

entanglement

?? Acting backwards ??

$$v = \begin{bmatrix} p \\ 1-p \end{bmatrix}$$

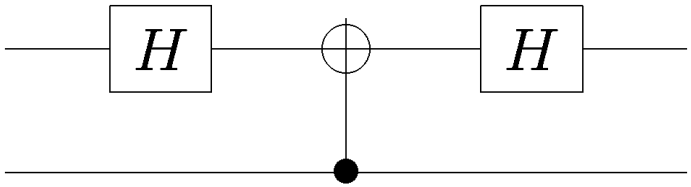
$$e = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{1}{2} \end{bmatrix}$$



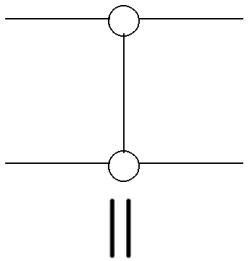

$$r = \begin{bmatrix} \frac{p}{2} \\ \frac{p}{2} \\ \frac{1-p}{2} \\ \frac{1-p}{2} \end{bmatrix}$$

$$v = \begin{bmatrix} p \\ 1-p \end{bmatrix}$$

# Deriving Quantum Teleportation



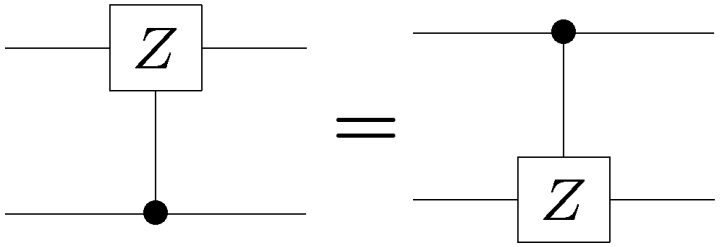

$=$



$=$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

$\|$

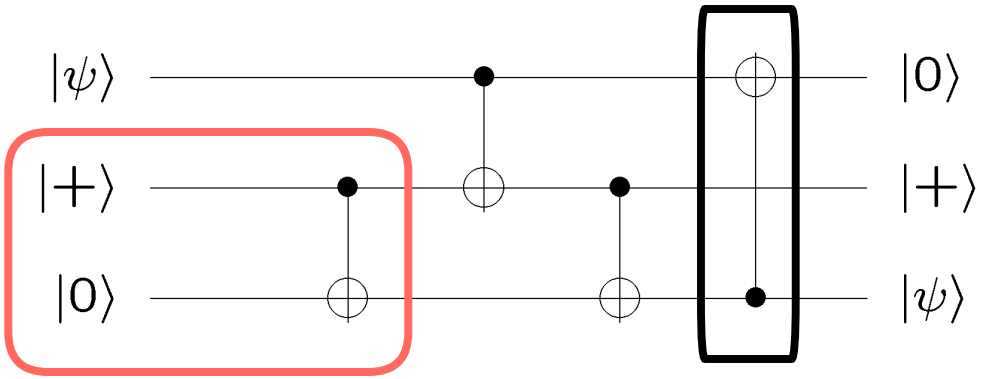



$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$HXH = Z$$

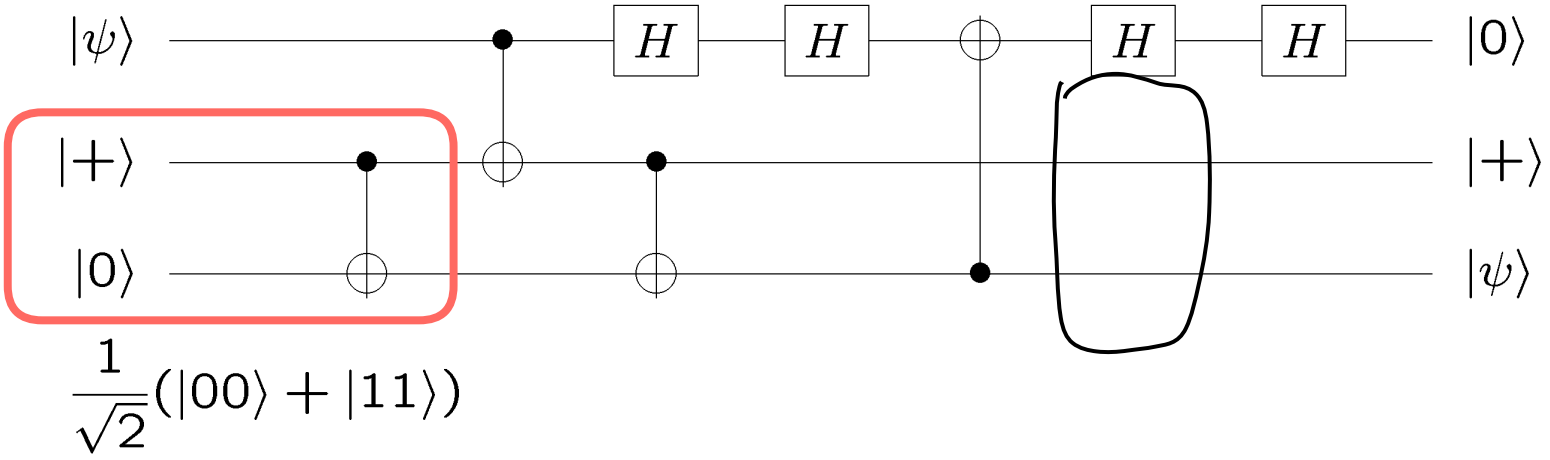
$$H^2 = I$$

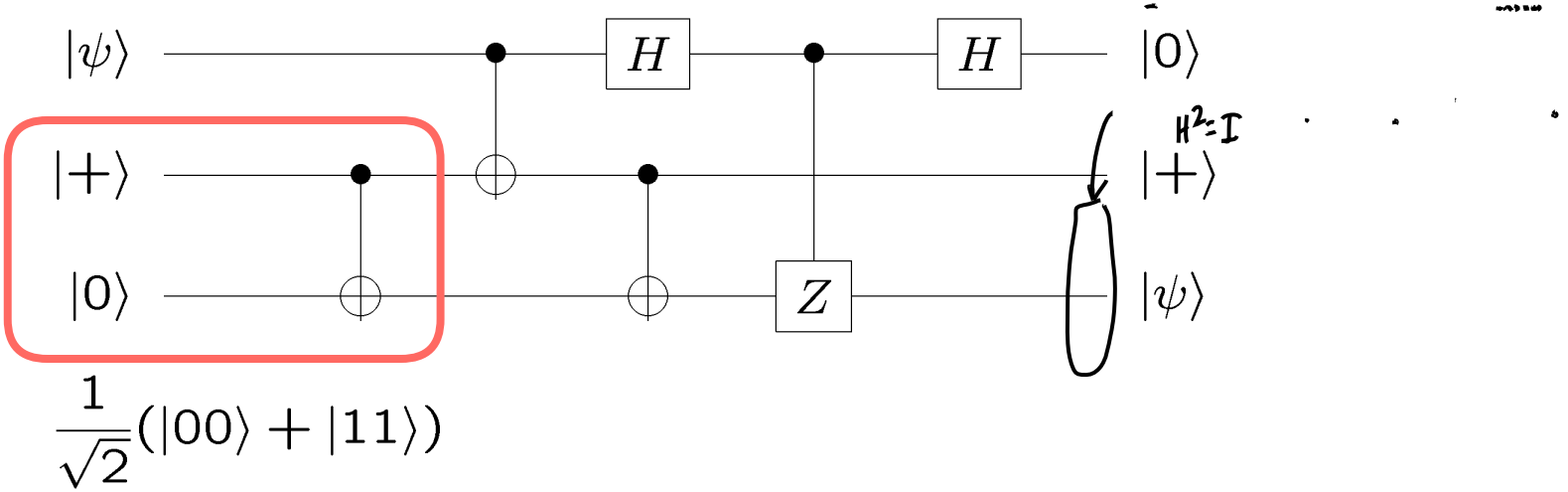
Use to turn around:



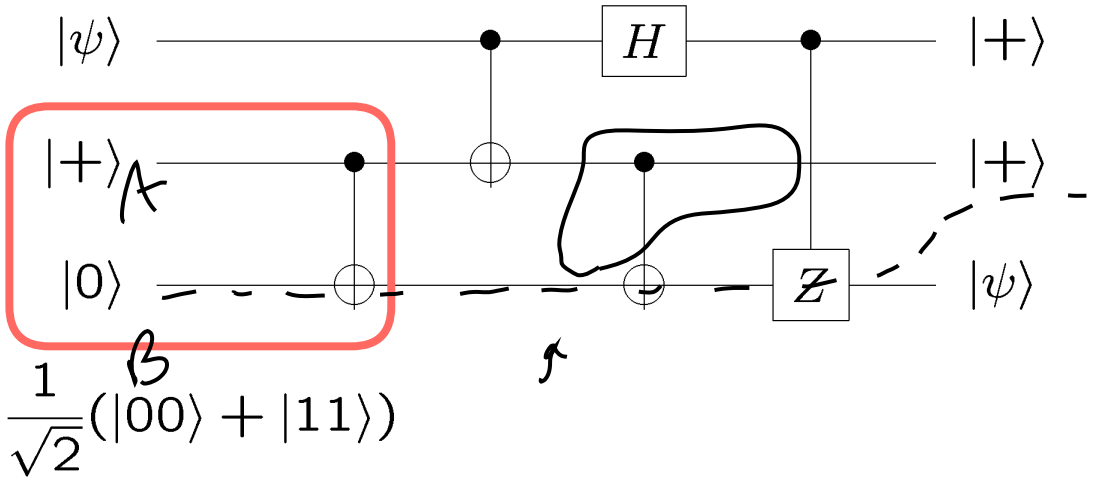

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

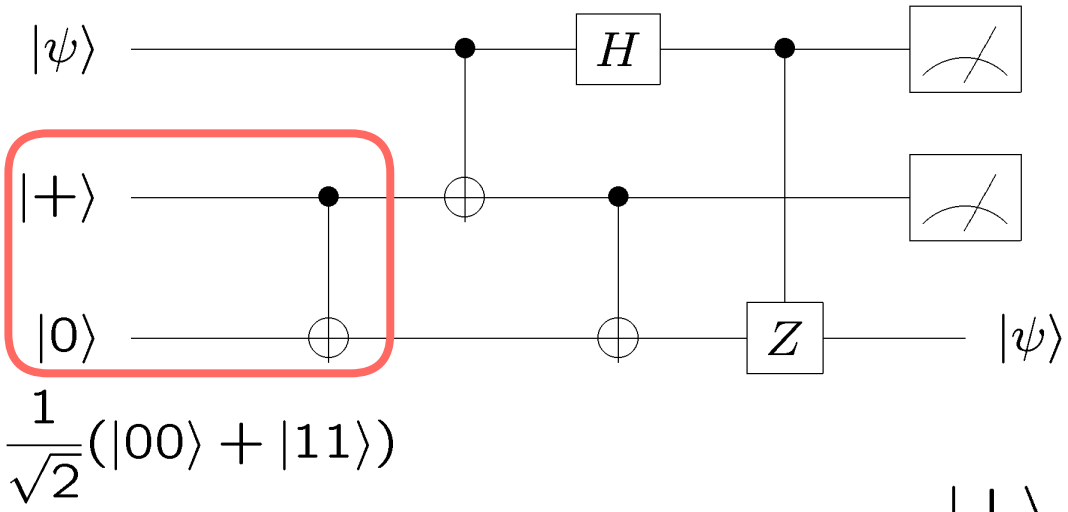
# Deriving Quantum Teleportation



# Deriving Quantum Teleportation



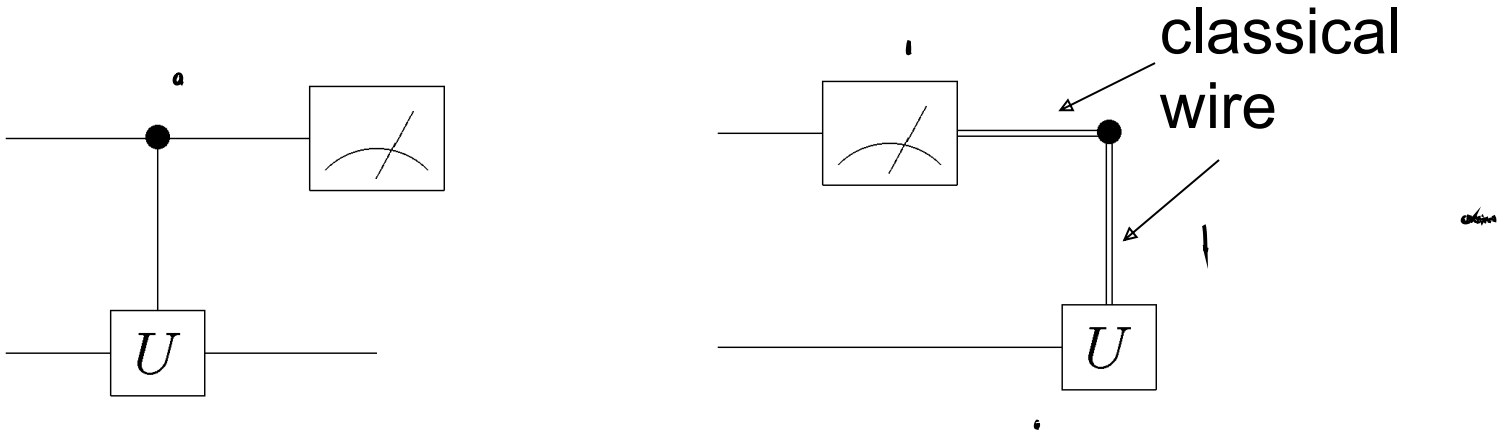

50 % 0, 50 % 1

50 % 0, 50 % 1

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

# Measurements Through Control

Measurement in the computational basis commutes with a control on a controlled unitary.



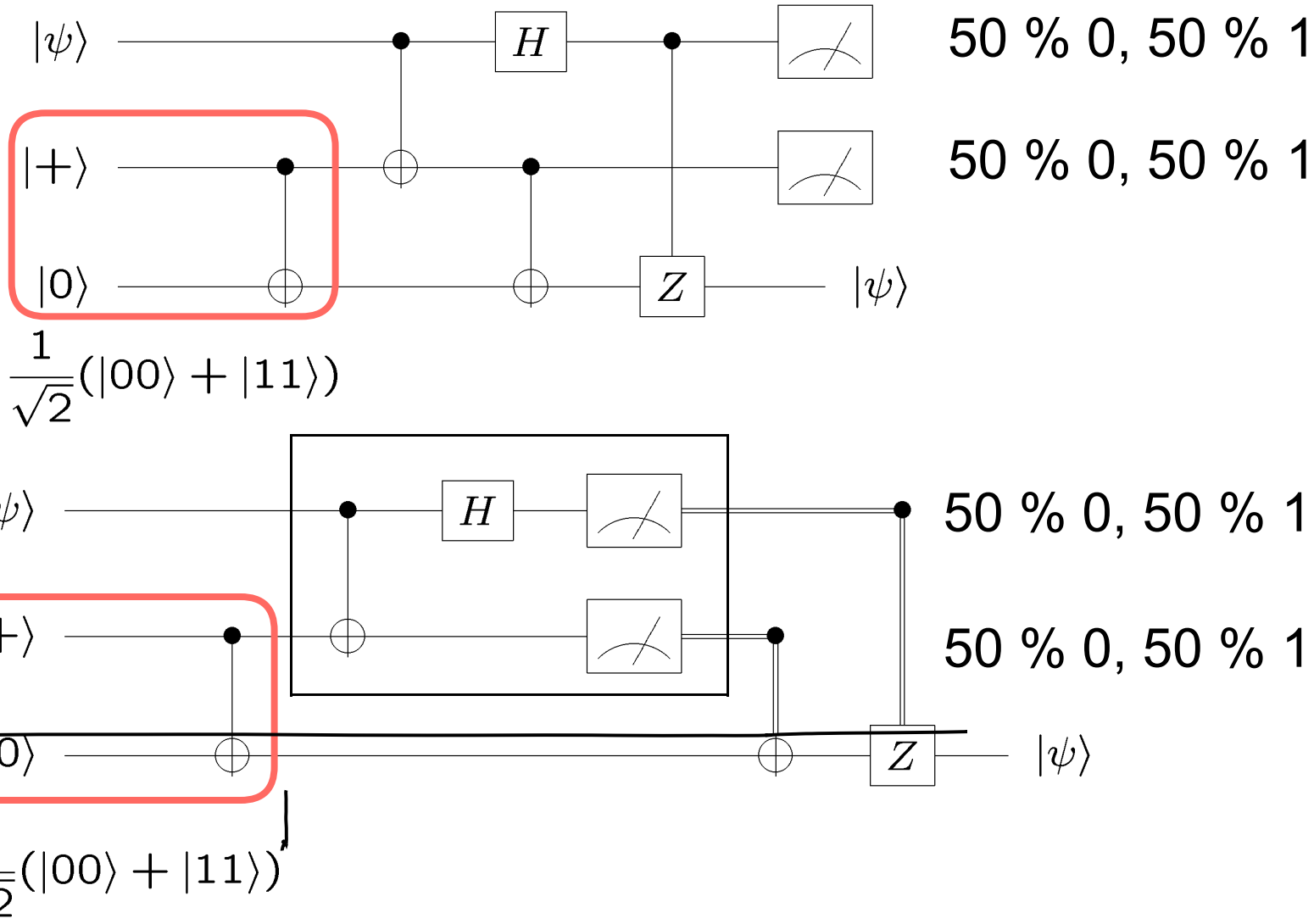

$$(P_0 \otimes I)C_U = C_U(P_0 \otimes I)$$

$$(|0\rangle\langle 0| \otimes I)(|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U) = (|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U)(|0\rangle\langle 0| \otimes I)$$

$$(P_1 \otimes I)C_U = C_U(P_1 \otimes I)$$

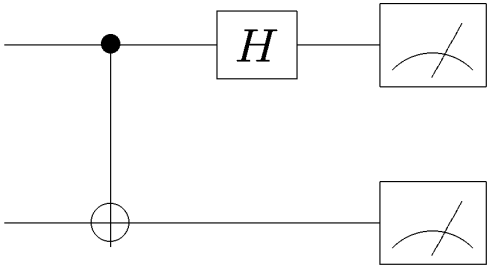
$$(|1\rangle\langle 1| \otimes I)(|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U) = (|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U)(|1\rangle\langle 1| \otimes I)$$

# Deriving Quantum Teleportation

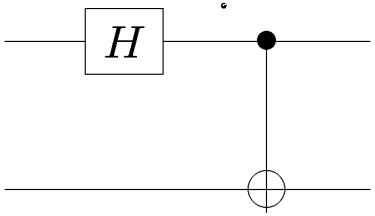


# Bell Basis Measurement

Unitary followed by measurement in the computational basis is a measurement in a different basis.




Run circuit backward to find basis:




$$|00\rangle \rightarrow \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

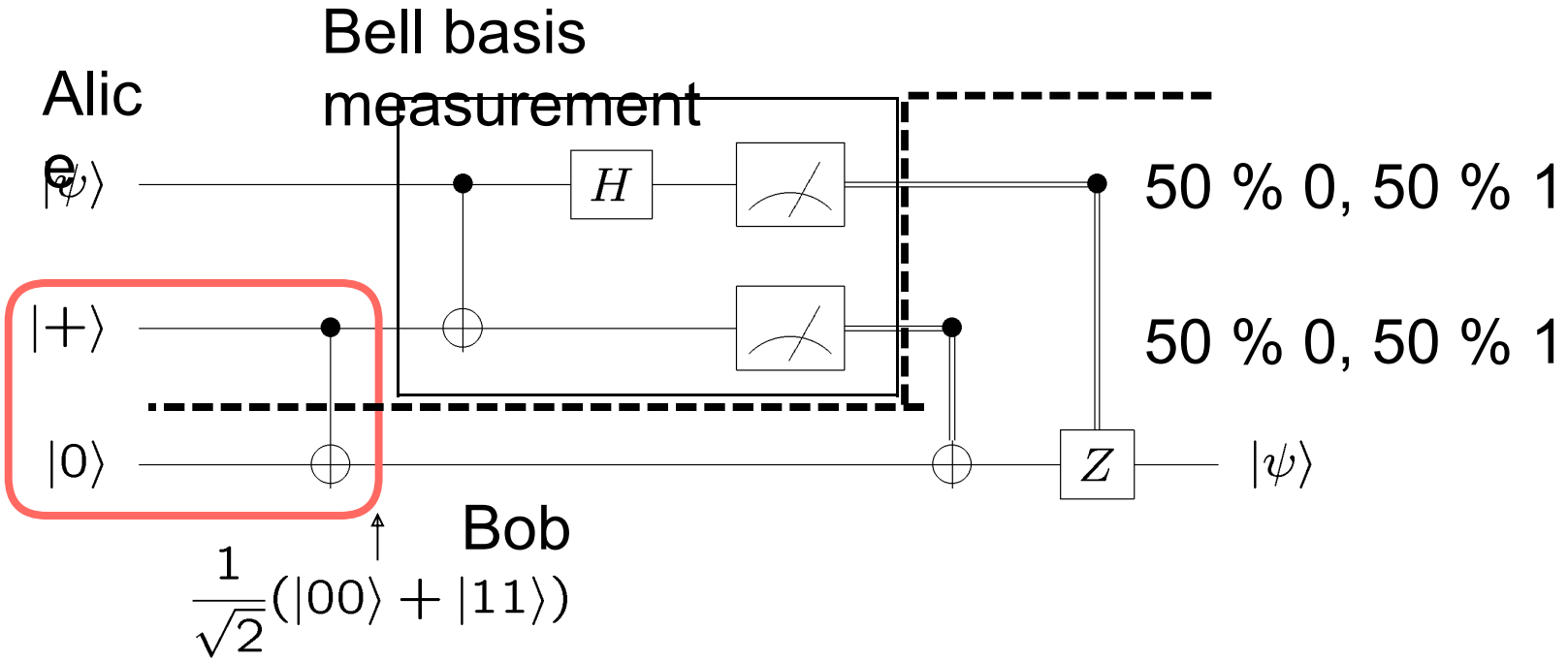
$$|01\rangle \rightarrow \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

$$|10\rangle \rightarrow \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$

$$|11\rangle \rightarrow \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

Thus we are measuring in the Bell basis.

# Teleportation




1. Initially Alice has $|\psi\rangle$ and they each have one of the two qubits of the entangled wave function $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
2. Alice measures $|\psi\rangle$ and her half of the entangled state in the Bell Basis.
3. Alice send the two bits of her outcome to Bob who then performs the appropriate X and Z operations to his qubit

# Teleportation

two qubits



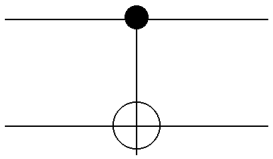
ALICE



BOB

*2.*
*Separate*
*1. Interact and entangle*

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$



$$|0\rangle$$

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Alice and Bob each have a qubit, and the wave function of their two qubit is entangled. This means that we can't think of Alice's qubit as having a particular wave function. We have to talk about the "global" two qubit wave function.

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \neq |a\rangle \otimes |b\rangle$$

# Teleportation



ALICE

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$



BOB

Alice does not know the wave function

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

We have three qubits whose wave function is

$$|\psi\rangle \otimes \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

qubit 1 — qubit 2 and qubit 3

# Separable, Entangled, 3 Qubits

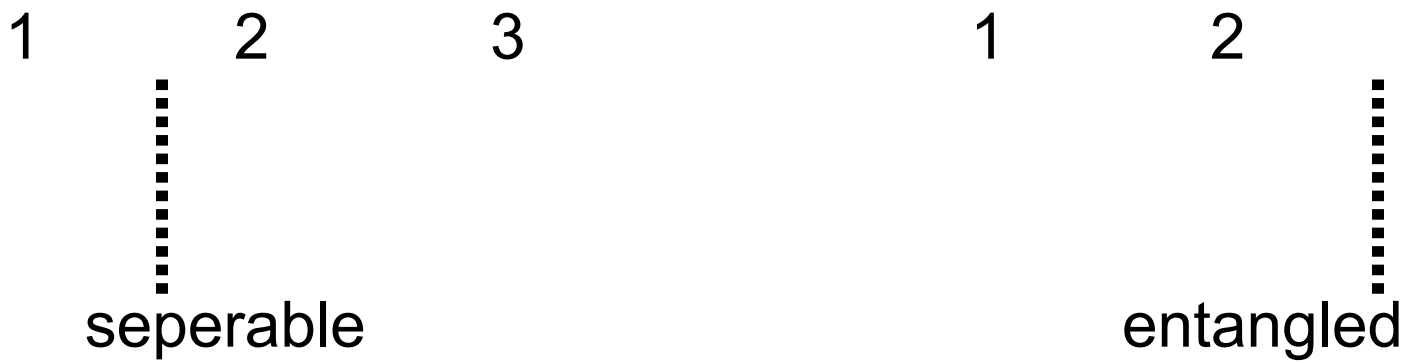
$$|\psi\rangle \otimes \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

If we consider qubit 1 as one subsystem and qubits 2 and 3 as another subsystem, then the state is separable across this divide

However, if we consider qubits 1 and 2 as one system and qubits 3 as one subsystem, then the state is entangled across this divide.

$$(\alpha|0\rangle+\beta|1\rangle)\otimes\frac{1}{\sqrt{2}}(|00\rangle+|11\rangle) = \frac{\alpha}{\sqrt{2}}|000\rangle+\frac{\alpha}{\sqrt{2}}|011\rangle+\frac{\beta}{\sqrt{2}}|100\rangle+\frac{\beta}{\sqrt{2}}|111\rangle$$

$$\neq (a_{00}|00\rangle+a_{01}|01\rangle+a_{10}|10\rangle+a_{11}|11\rangle)\otimes(b_0|0\rangle+b_1|1\rangle)$$



# Separable, Entangled, 3 Qubits

Sometimes we will deal with entangled states across non adjacent qubits:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

How do we even "write" this? $|\psi\rangle_2 \otimes \frac{1}{\sqrt{2}}(|00\rangle_{13} + |11\rangle_{13})$

Subscript denotes which qubit(s) you are talking about.

# Separable, Entangled, 3 Qubits

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

1 2 3

$$|\psi\rangle_1 \otimes \frac{1}{\sqrt{2}}(|00\rangle_{23} + |11\rangle_{23})$$

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

1 2 3

$$|\psi\rangle_2 \otimes \frac{1}{\sqrt{2}}(|00\rangle_{13} + |11\rangle_{13})$$

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

# Separable, Entangled, 3 Qubits

$$(\alpha|0\rangle_1 + \beta|1\rangle_1 \otimes \frac{1}{\sqrt{2}}(|00\rangle_{23} + |11\rangle_{23})$$

$$= \frac{\alpha}{\sqrt{2}}|000\rangle_{123} + \frac{\alpha}{\sqrt{2}}|011\rangle_{123} + \frac{\beta}{\sqrt{2}}|100\rangle_{123} + \frac{\beta}{\sqrt{2}}|111\rangle_{123}$$

$$= \frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|100\rangle + \frac{\beta}{\sqrt{2}}|111\rangle$$

$$(\alpha|0\rangle_2 + \beta|1\rangle_2) \otimes \frac{1}{\sqrt{2}}(|00\rangle_{13} + |11\rangle_{13})$$

$$= \frac{\alpha}{\sqrt{2}}|000\rangle_{213} + \frac{\alpha}{\sqrt{2}}|011\rangle_{213} + \frac{\beta}{\sqrt{2}}|100\rangle_{213} + \frac{\beta}{\sqrt{2}}|111\rangle_{213}$$

$$= \frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|101\rangle + \frac{\beta}{\sqrt{2}}|010\rangle + \frac{\beta}{\sqrt{2}}|111\rangle$$

When we don't write subscripts we mean "standard ordering"

# Teleportation



ALICE

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$



BOB

Alice does not know the wave function

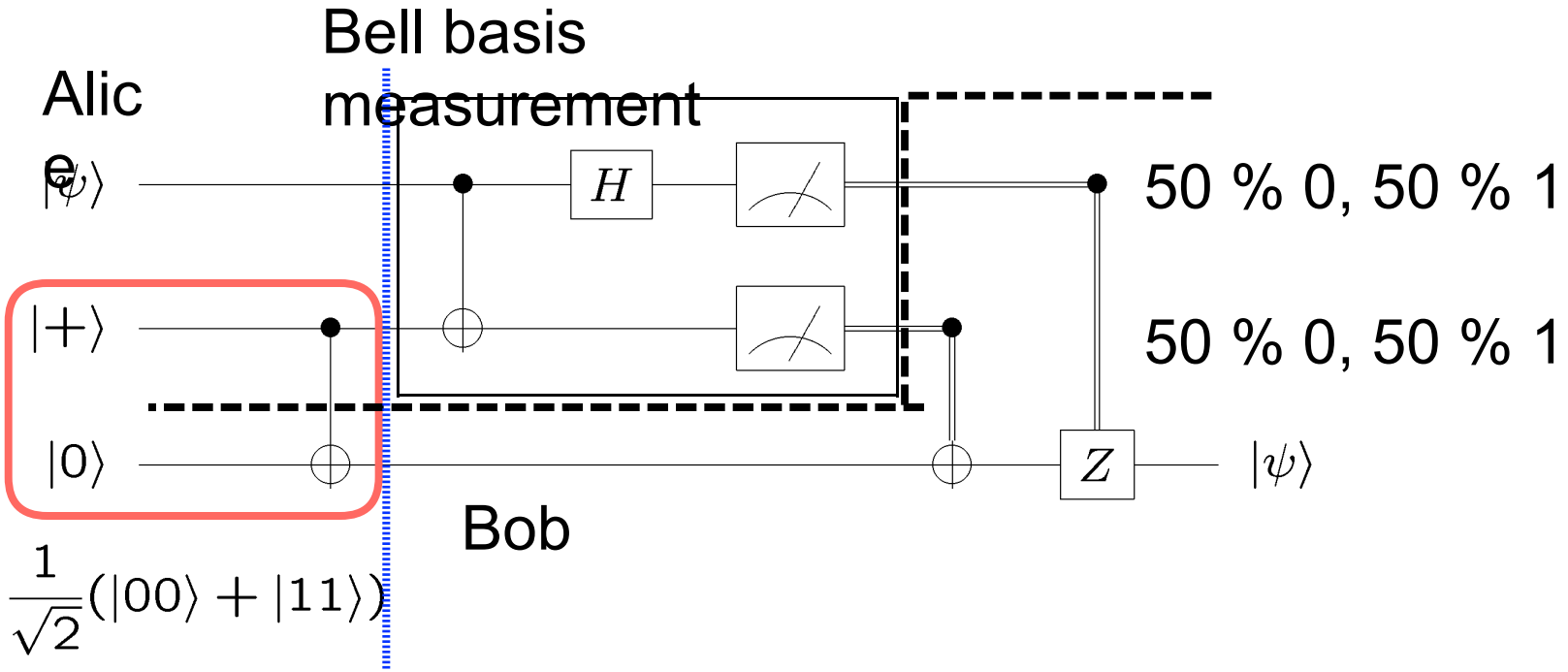
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

We have three qubits whose wave function is

$$|\psi\rangle \otimes \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

qubit 1 — qubit 2 and qubit 3

# Teleportation




$$|\psi\rangle_1 \otimes \frac{1}{\sqrt{2}}(|00\rangle_{23} + |11\rangle_{23})$$

$$= \frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|100\rangle + \frac{\beta}{\sqrt{2}}|111\rangle$$

# Teleportation

$$|\psi\rangle_1 \otimes \frac{1}{\sqrt{2}}(|00\rangle_{23} + |11\rangle_{23})$$

$$= \frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|100\rangle + \frac{\beta}{\sqrt{2}}|111\rangle$$

Express this state in terms of Bell basis for first two qubits.

Bell basis

$$|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$|\Phi_-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$

$$|\Psi_+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

$$|\Psi_-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

Computational basis

$$|00\rangle = \frac{1}{\sqrt{2}}(|\Phi_+\rangle + |\Phi_-\rangle)$$

$$|01\rangle = \frac{1}{\sqrt{2}}(|\Psi_+\rangle + |\Psi_-\rangle)$$

$$|10\rangle = \frac{1}{\sqrt{2}}(|\Phi_+\rangle - |\Phi_-\rangle)$$

$$|01\rangle = \frac{1}{\sqrt{2}}(|\Phi_+\rangle - |\Phi_-\rangle)$$

# Teleportation

Bell basis

$$|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$
$$|\Phi_-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$
$$|\Psi_+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$
$$|\Psi_-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

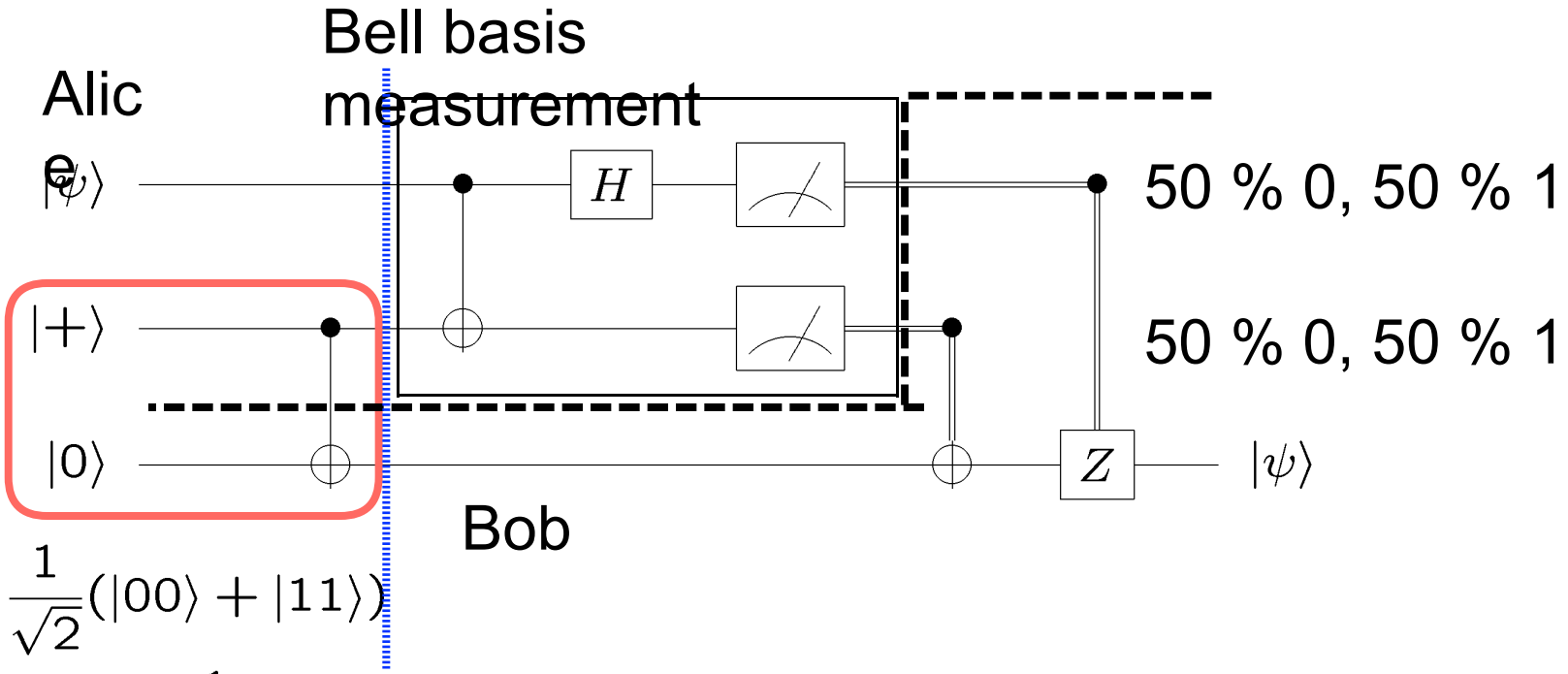
Computational basis

$$|00\rangle = \frac{1}{\sqrt{2}}(|\Phi_+\rangle + |\Phi_-\rangle)$$
$$|01\rangle = \frac{1}{\sqrt{2}}(|\Psi_+\rangle + |\Psi_-\rangle)$$
$$|11\rangle = \frac{1}{\sqrt{2}}(|\Phi_+\rangle - |\Phi_-\rangle)$$
$$|10\rangle = \frac{1}{\sqrt{2}}(|\Psi_+\rangle - |\Psi_-\rangle)$$

$$\frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|100\rangle + \frac{\beta}{\sqrt{2}}|111\rangle$$

$$\begin{aligned} &= \frac{1}{2}(\alpha(|\Phi_+\rangle + |\Phi_-\rangle) \otimes |0\rangle + \alpha(|\Psi_+\rangle + |\Psi_-\rangle) \otimes |1\rangle \\ &\quad +\beta(|\Psi_+\rangle - |\Psi_-\rangle) \otimes |0\rangle + \beta(|\Phi_+\rangle - |\Phi_-\rangle) \otimes |1\rangle \\ &= \frac{1}{2}\Big[|\Phi_+\rangle \otimes (\alpha|0\rangle + \beta|1\rangle) + |\Phi_-\rangle \otimes (\alpha|0\rangle - \beta|1\rangle) \\ &\quad +|\Psi_+\rangle \otimes (\alpha|1\rangle + \beta|0\rangle) + |\Psi_-\rangle \otimes (\alpha|1\rangle - \beta|0\rangle)\Big] \end{aligned}$$

# Teleportation




$$|\psi\rangle_1 \otimes \frac{1}{\sqrt{2}}(|00\rangle_{23} + |11\rangle_{23})$$

$$= \frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|100\rangle + \frac{\beta}{\sqrt{2}}|111\rangle$$

$$= \frac{1}{2}\Big[|\Phi_+\rangle_{12}(\alpha|0\rangle_3 + \beta|1\rangle_3) + |\Phi_-\rangle_{12}(\alpha|0\rangle_3 - \beta|1\rangle_3)$$

$$+|\Psi_+\rangle_{12}(\alpha|1\rangle_3 + \beta|0\rangle_3) + |\Psi_-\rangle_{12}(\alpha|1\rangle_3 - \beta|0\rangle_3)\Big]$$

# Dropping The Tensor Symbol

$$\frac{1}{2}\Big[|\Phi_+\rangle \otimes (\alpha|0\rangle + \beta|1\rangle + |\Phi_-\rangle \otimes (\alpha|0\rangle - \beta|1\rangle)$$
$$+|\Psi_+\rangle \otimes (\alpha|1\rangle + \beta|0\rangle) + |\Psi_-\rangle \otimes (\alpha|1\rangle - \beta|0\rangle)\Big]$$

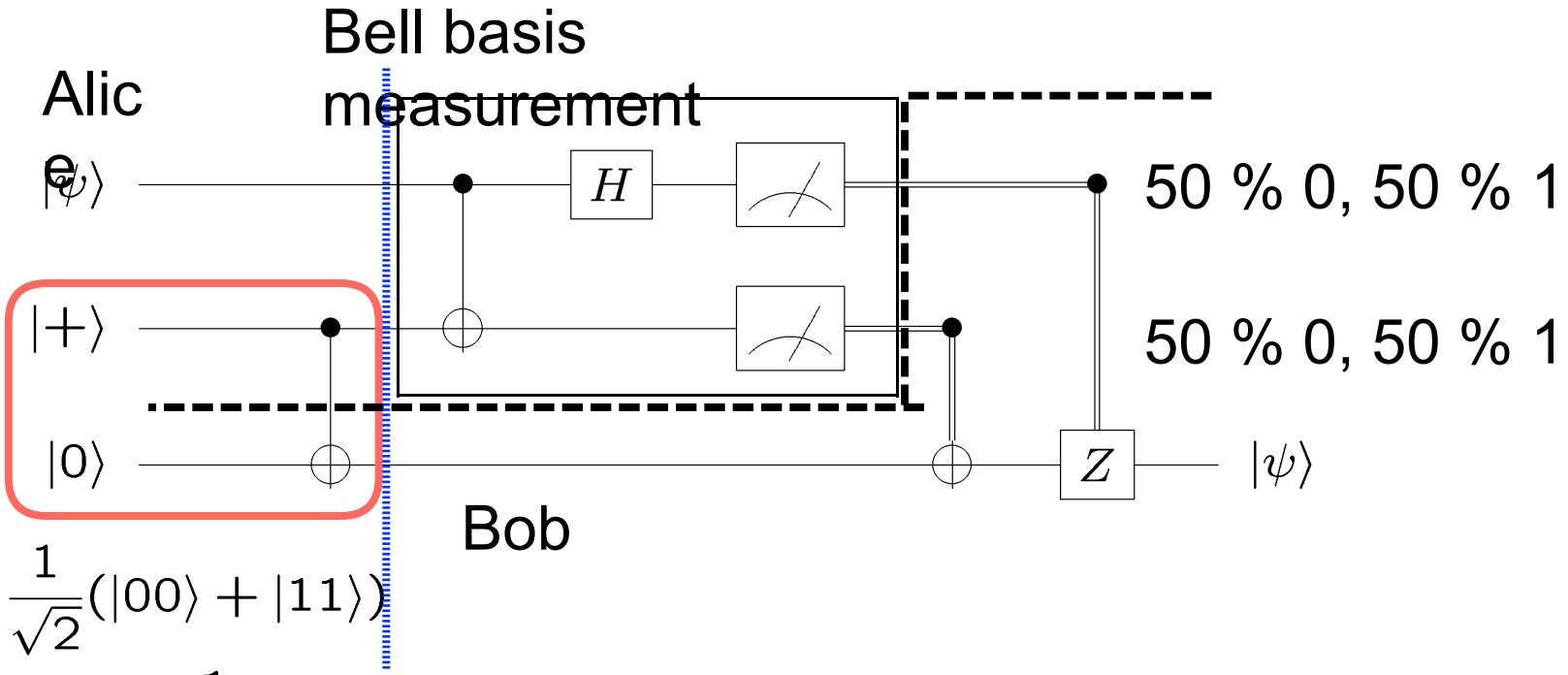
$$= \frac{1}{2}\Big[|\Phi_+\rangle_{12}(\alpha|0\rangle_3 + \beta|1\rangle_3) + |\Phi_-\rangle_{12}(\alpha|0\rangle_3 - \beta|1\rangle_3)$$
$$+|\Psi_+\rangle_{12}(\alpha|1\rangle_3 + \beta|0\rangle_3) + |\Psi_-\rangle_{12}(\alpha|1\rangle_3 - \beta|0\rangle_3)\Big]$$

$$= \frac{1}{2}\Big[|\Phi_+\rangle(\alpha|0\rangle) + \beta|1\rangle) + |\Phi_-\rangle(\alpha|0\rangle - \beta|1\rangle)$$
$$+|\Psi_+\rangle(\alpha|1\rangle + \beta|0\rangle) + |\Psi_-\rangle(\alpha|1\rangle - \beta|0\rangle)\Big]$$

Sometimes we will just “drop” the tensor symbol.

“Context” lets us know that there is an implicit tensor product.

# Teleportation



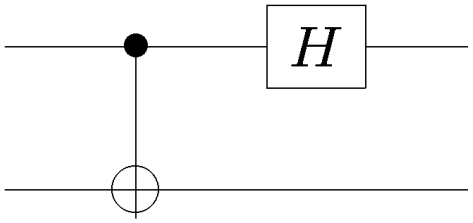

$$|\psi\rangle_1 \otimes \frac{1}{\sqrt{2}}(|00\rangle_{23} + |11\rangle_{23})$$

$$= \frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|100\rangle + \frac{\beta}{\sqrt{2}}|111\rangle$$

$$= \frac{1}{2}\Big[|\Phi_+\rangle_{12}(\alpha|0\rangle_3) + \beta|1\rangle_3) + |\Phi_-\rangle_{12}(\alpha|0\rangle_3 - \beta|1\rangle_3)$$

$$+|\Psi_+\rangle_{12}(\alpha|1\rangle_3 + \beta|0\rangle_3) + |\Psi_-\rangle_{12}(\alpha|1\rangle_3 - \beta|0\rangle_3)\Big]$$

# Bell Basis Measurement




$$|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \rightarrow |00\rangle \quad |\Phi_-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \rightarrow |10\rangle$$

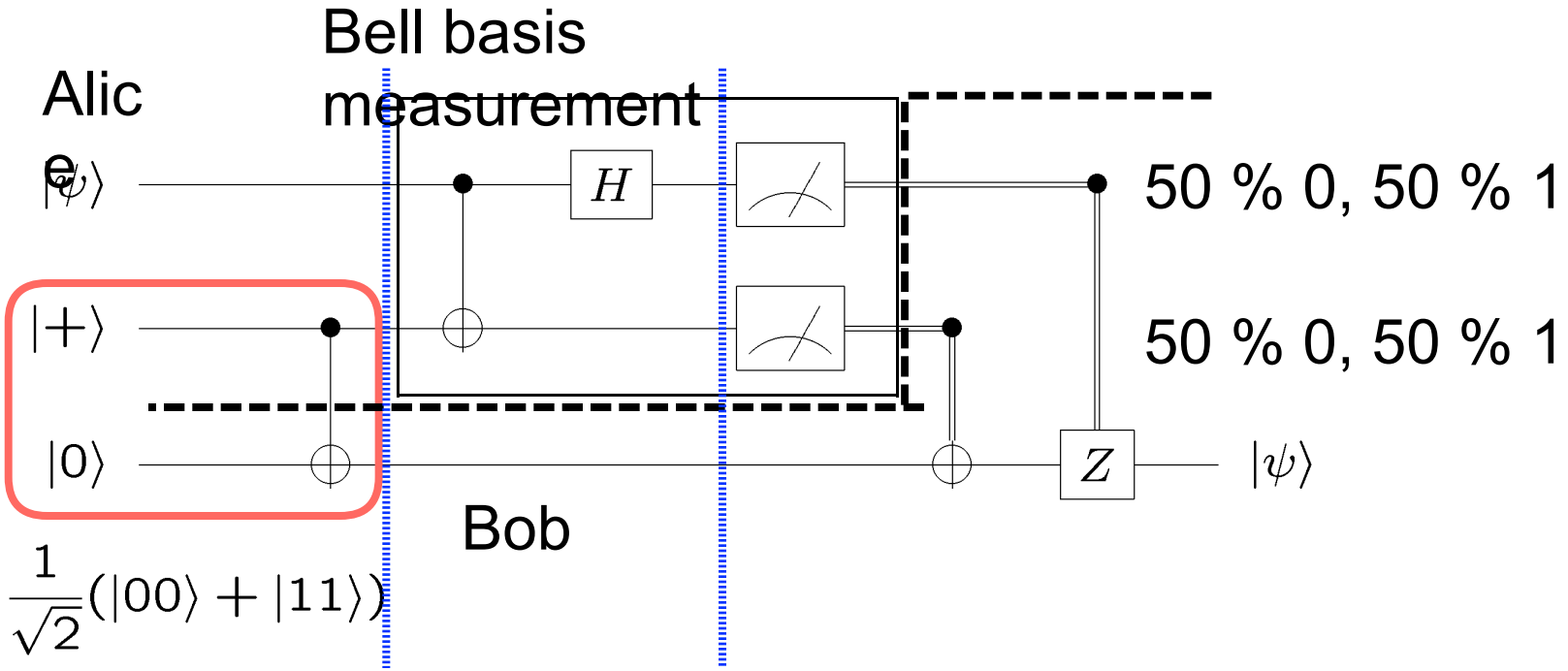
$$|\Psi_+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \rightarrow |01\rangle \quad |\Psi_-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \rightarrow |11\rangle$$

$$\frac{1}{2}\Big[|\Phi_+\rangle_{12}(\alpha|0\rangle_3 + \beta|1\rangle_3) + |\Phi_-\rangle_{12}(\alpha|0\rangle_3 - \beta|1\rangle_3)$$
$$+|\Psi_+\rangle_{12}(\alpha|1\rangle_3 + \beta|0\rangle_3) + |\Psi_-\rangle_{12}(\alpha|1\rangle_3 - \beta|0\rangle_3)\Big]$$

$$\downarrow$$

$$\frac{1}{2}[|00\rangle_{12}(\alpha|0\rangle_3 + \beta|1\rangle_3) + |10\rangle_{12}(\alpha|0\rangle_3 - \beta|1\rangle_3)$$
$$+|01\rangle_{12}(\alpha|1\rangle_3 + \beta|0\rangle_3) + |11\rangle_{12}(\alpha|1\rangle_3 - \beta|0\rangle_3)]$$

# Teleportation




$$\frac{1}{2}\Big[|\Phi_+\rangle_{12}(\alpha|0\rangle_3 + \beta|1\rangle_3) + |\Phi_-\rangle_{12}(\alpha|0\rangle_3 - \beta|1\rangle_3)$$
$$+|\Psi_+\rangle_{12}(\alpha|1\rangle_3 + \beta|0\rangle_3) + |\Psi_-\rangle_{12}(\alpha|1\rangle_3 - \beta|0\rangle_3)\Big]$$

$$\frac{1}{2}[|00\rangle_{12}(\alpha|0\rangle_3 + \beta|1\rangle_3) + |10\rangle_{12}(\alpha|0\rangle_3 - \beta|1\rangle_3)$$
$$+|01\rangle_{12}(\alpha|1\rangle_3 + \beta|0\rangle_3) + |11\rangle_{12}(\alpha|1\rangle_3 - \beta|0\rangle_3)]$$

# Teleportation

Given the wave fun

$$\frac{1}{2}[|00\rangle_{12}(\alpha|0\rangle_3+\beta|1\rangle_3)+|10\rangle_{12}(\alpha|0\rangle_3-\beta|1\rangle_3)$$
$$+|01\rangle_{12}(\alpha|1\rangle_3+\beta|0\rangle_3)+|11\rangle_{12}(\alpha|1\rangle_3-\beta|0\rangle_3)]$$

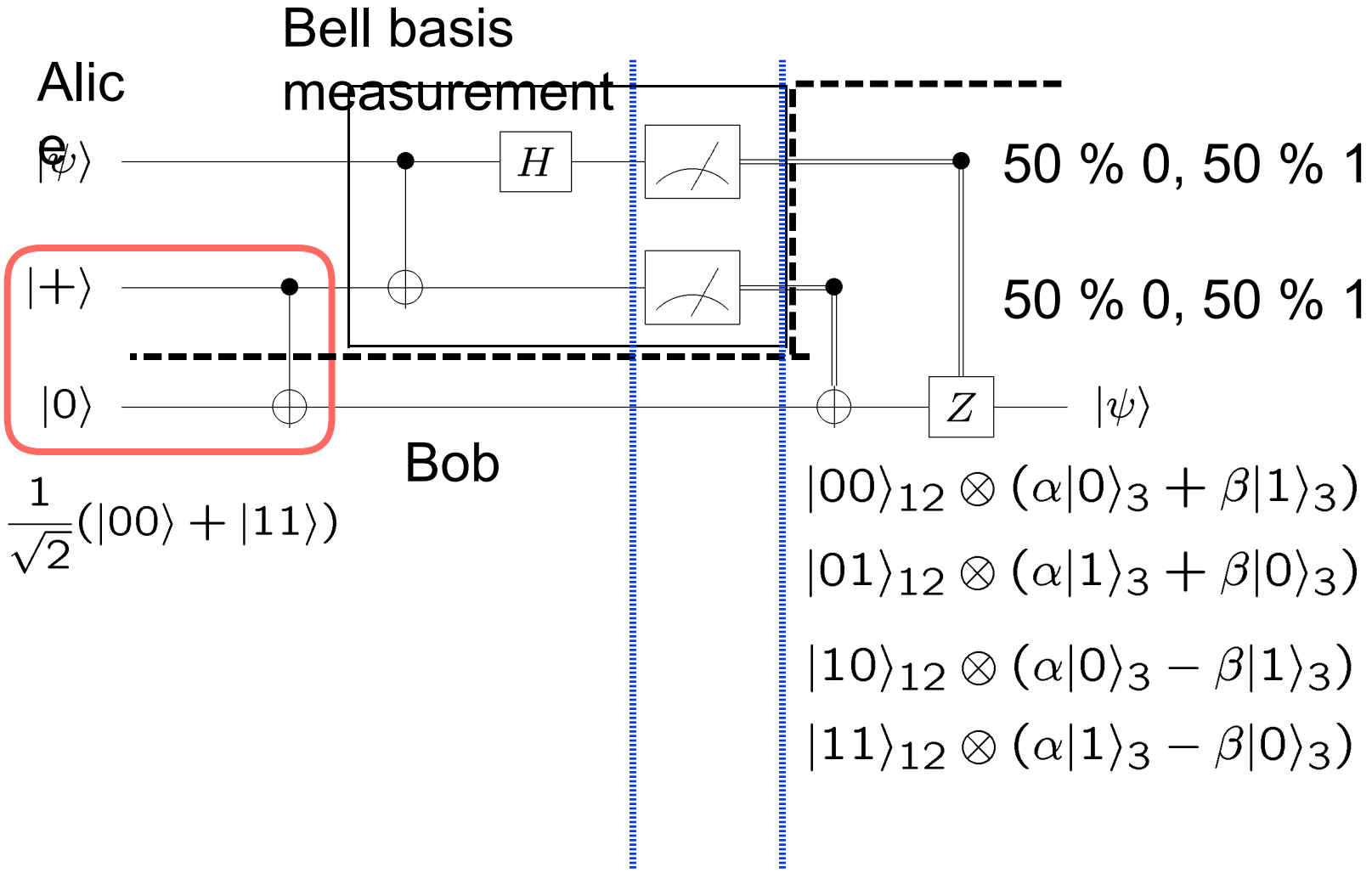
Measure the first two qubits in the computational basis

$$M_{00}=|00\rangle\langle 00|\otimes I \qquad M_{01}=|01\rangle\langle 01|\otimes I$$
$$M_{10}=|10\rangle\langle 10|\otimes I \qquad M_{11}=|11\rangle\langle 11|\otimes I$$

Equal ¼ probability for all four outcomes and new states are:

$$|00\rangle_{12}\otimes(\alpha|0\rangle_3+\beta|1\rangle_3)$$
$$|01\rangle_{12}\otimes(\alpha|1\rangle_3+\beta|0\rangle_3)$$
$$|10\rangle_{12}\otimes(\alpha|0\rangle_3-\beta|1\rangle_3)$$
$$|11\rangle_{12}\otimes(\alpha|1\rangle_3-\beta|0\rangle_3)$$

# Teleportation




$$\frac{1}{2}[|00\rangle_{12}(\alpha|0\rangle_3 + \beta|1\rangle_3) + |10\rangle_{12}(\alpha|0\rangle_3 - \beta|1\rangle_3)$$
$$+|01\rangle_{12}(\alpha|1\rangle_3 + \beta|0\rangle_3) + |11\rangle_{12}(\alpha|1\rangle_3 - \beta|0\rangle_3)]$$

# Teleportation

If the bits sent from Alice to Bob are 00, do nothing

$$|00\rangle_{12} \otimes (\alpha|0\rangle_3 + \beta|1\rangle_3) = |00\rangle_{12} \otimes |\psi\rangle_3$$

If the bits sent from Alice to Bob are 01, apply a bit flip

$$(I_4 \otimes X)|01\rangle_{12} \otimes (\alpha|1\rangle_3 + \beta|0\rangle_3) = |01\rangle_{12} \otimes (\alpha|0\rangle_3 + \beta_1|1\rangle_3)$$

$$= |01\rangle_{12} \otimes |\psi\rangle_3$$

If the bits sent from Alice to Bob are 10, apply a phase flip

$$(I_4 \otimes Z)|10\rangle_{12} \otimes (\alpha|0\rangle_3 - \beta|1\rangle_3) = |10\rangle \otimes (\alpha|0\rangle_3 + \beta|1\rangle_3)$$

$$= |10\rangle_{12} \otimes |\psi\rangle_3$$

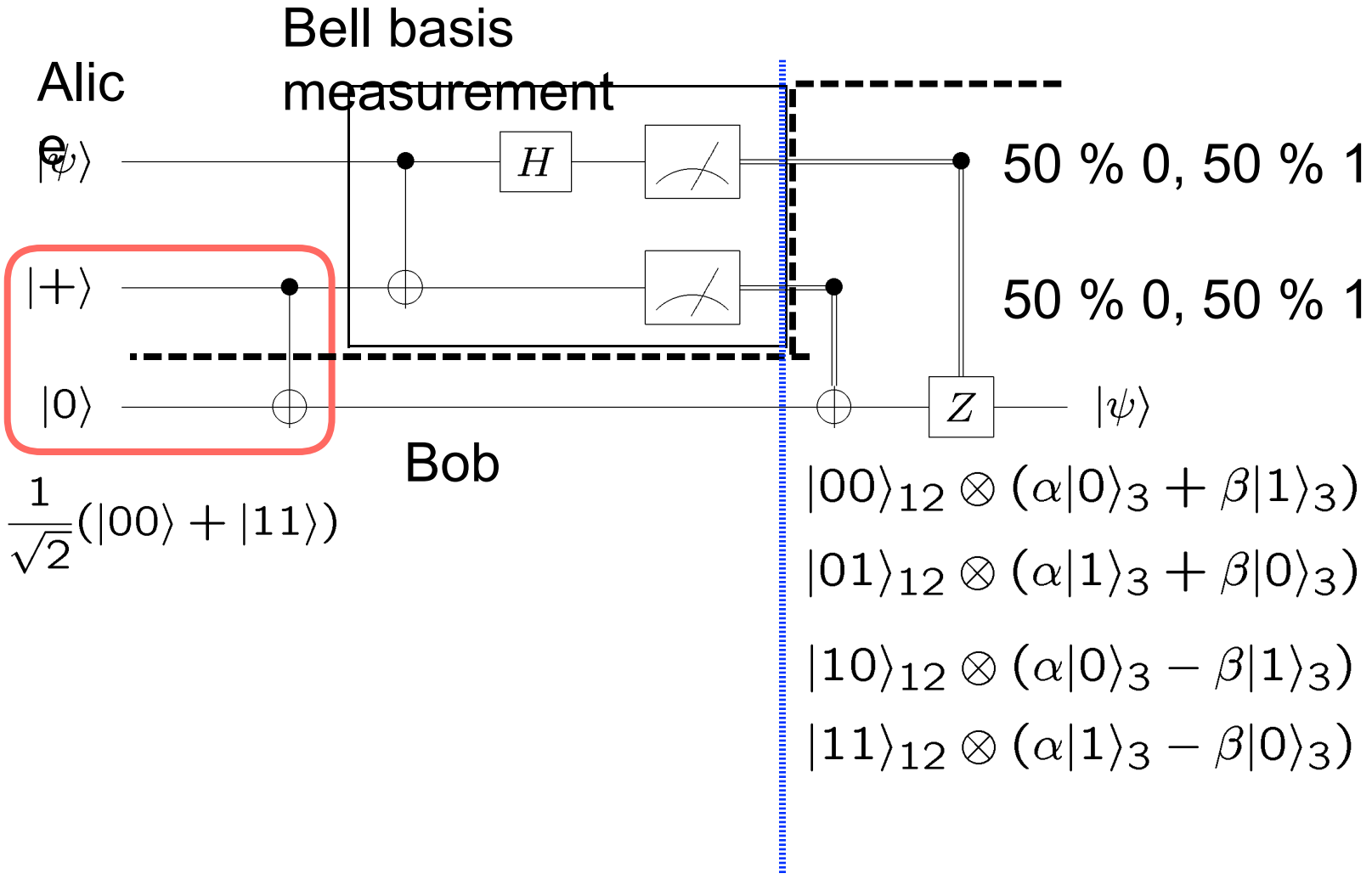
If the bits sent from Alice to Bob are 11, apply a bit & phase flip

$$(I_4 \otimes Z)(I_4 \otimes X)|11\rangle_{12} \otimes (\alpha|1\rangle_3 - \beta|0\rangle_3) = (I_4 \otimes Z)|11\rangle_{12} \otimes (\alpha|0\rangle_3 - \beta|1\rangle_3)$$

$$= |11\rangle_{12} \otimes (\alpha|0\rangle_3 + \beta|1\rangle_3)$$

$$= |11\rangle_{12} \otimes |\psi\rangle_3$$

# Teleportation



# Teleportation



Alice



Bob

$\cong$

$|\psi\rangle$

Pauli

2 bits

Bell measurement

$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$



$|\psi\rangle$

Alice



Bob

# Teleportation

1 qubit = 1 ebit + 2 bits

Teleportation says we can replace transmitting a qubit with a shared entangled pair of qubits plus two bits of classical communication.

# Superdense Coding

Next we will see that

2 bits = 1 qubit + 1 ebit

# Superdense Coding

Suppose Alice and Bob each have one qubit and the joint two qubit wave function is the entangled state

$$|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Alice wants to send two bits to Bob. Call these bits $b_1$ and $b_2$

Alice applies the following operator to her qubit: $Z^{b_2}X^{b_1}$

Alice then sends her qubit to Bob.

note: $M^0 = I$

Bob then measures in the Bell basis to determine the two bits

| 2 bits = 1 qubit + 1 ebit |
|---|

# Bell Basis

The four Bell states can be turned into each other using operations on only one of the qubits:

$$|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$(X\otimes I)|\Phi_+\rangle = (X\otimes I)\frac{1}{\sqrt{2}}(|00\rangle+|11\rangle) = \frac{1}{\sqrt{2}}(|10\rangle+|01\rangle) = |\Psi_+\rangle$$

$$(Z\otimes I)|\Phi_+\rangle = (Z\otimes I)\frac{1}{\sqrt{2}}(|00\rangle+|11\rangle) = \frac{1}{\sqrt{2}}(|00\rangle-|11\rangle) = |\Phi_-\rangle$$

$$(ZX\otimes I)|\Phi_+\rangle = (ZX\otimes I)\frac{1}{\sqrt{2}}(|00\rangle+|11\rangle) = \frac{1}{\sqrt{2}}(-|10\rangle+|01\rangle) = |\Psi_-\rangle$$

# Superdense Coding

Initially $|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

:

Alice applies the following operator to her qubit: $Z^{b_2}X^{b_1}$

$$(Z^{b_2}X^{b_1} \otimes I)|\Phi_+\rangle$$

$$\begin{aligned}
b_1 = 0, b_2 = 0 &\qquad |\Phi_+\rangle \\
b_1 = 0, b_2 = 1 &\qquad (Z \otimes I)|\Phi_+\rangle = |\Phi_-\rangle \\
b_1 = 1, b_2 = 0 &\qquad (X \otimes I)|\Phi_+\rangle = |\Psi_+\rangle \\
b_1 = 1, b_2 = 1 &\qquad (ZX \otimes I)|\Phi_+\rangle = |\Psi_-\rangle
\end{aligned}$$

Bob can uniquely determine which of the four states he has and thus figure out Alice's two bits!

# Superdense Coding



$b_1$

$b_2$

$Z^{b_2} X^{b_1}$

$|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$



$b_1$

$b_2$

Bell basis measurement

# Teleportation

1 qubit = 1 ebit + 2 bits

Teleportation says we can replace transmitting a qubit with a shared entangled pair of qubits plus two bits of classical communication.

# Superdense Coding

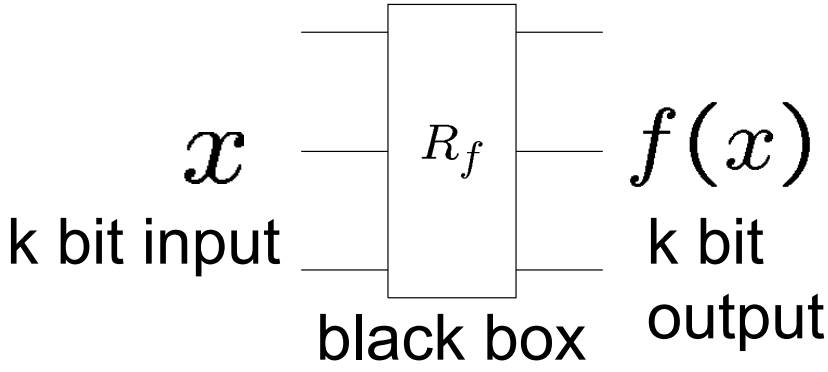
2 bits = 1 qubit + 1 ebit

We can send two bits of classical information if we share an entangled state and can communicate one qubit of quantum information:

# Quantum Algorithms

# Classical Promise Problem Query Complexity

**Given:** A black box which computes some function



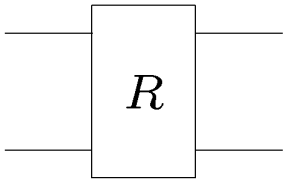

**Promise:** the function belongs to a set $\mathcal{S}$ which is a subset of all possible functions.

**Properties:** the set $\mathcal{S}$ can be divided into disjoint subsets $\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_m$

**Problem:** What is the minimal number of times we have to use (query) the black box in order to determine which subset $\mathcal{S}_i$ the function belongs to?

# Example

Suppose you are given a black box which computes one of the following four reversible classical gates:




2 bits input

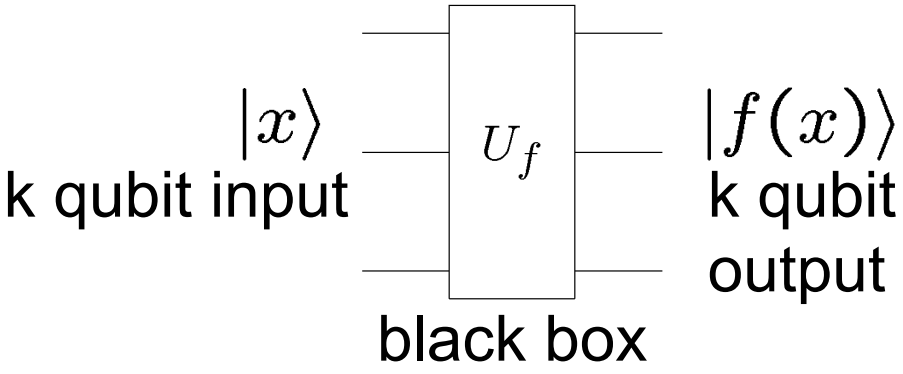
2 bits output

| | | | |
|---|---|---|---|
| $00 \to 00$ | $00 \to 01$ | $00 \to 00$ | $00 \to 01$ |
| $01 \to 01$ | $01 \to 00$ | $01 \to 01$ | $01 \to 00$ |
| $10 \to 10$ | $10 \to 11$ | $10 \to 11$ | $10 \to 10$ |
| $11 \to 11$ | $11 \to 10$ | $11 \to 10$ | $11 \to 11$ |
| "identity" | NOT 2nd bit | controlled-NOT | controlled-NOT + NOT 2nd bit |
| | $\mathcal{S}_1$ | | $\mathcal{S}_2$ |

**Deutsch's (Classical) Problem:** What is the minimal number of times we have to use this black box to determine whether we are given one of the first two or the second two functions?

# Quantum Promise Query Complexity

**Given:** A quantum gate which, when used as a classical device computes a reversible function



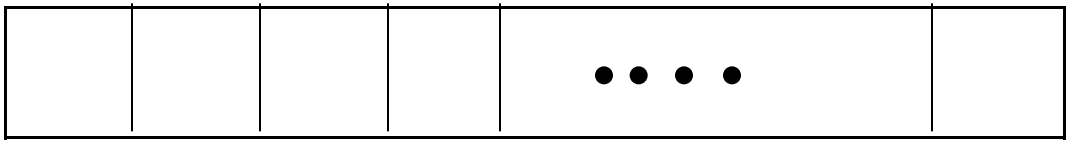

**Promise:** the function belongs to a set $\mathcal{S}$ which is a subset of all possible functions.

**Properties:** the set $\mathcal{S}$ can be divided into disjoint subsets $\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_m$

**Problem:** What is the minimal number of times we have to use (query) the quantum gate in order to determine which subset $\mathcal{S}_i$ the function belongs to?

# n Qubit Registers

Up until now, we have dealt with only 1,2,3, or 4 qubits.
Now we will deal with n qubits at a time!



n
qubits

Computational basis:

$$|x_1\rangle \otimes |x_2\rangle \otimes |x_3\rangle \otimes \cdots \otimes |x_n\rangle \qquad x_i \in \{0,1\}$$

$$|x_1, x_2, x_3, \ldots, x_n\rangle$$

$|x\rangle \quad x \in \{0,1\}^n$ n bit string

$$|x\rangle \quad x \in \{0,1,2,\ldots,2^n-1\}$$

# n Qubit States

n qubits have a wave function with $2^n$ complex numbers.

Writing $2^n$ complex numbers down, and keeping track of them (in a naïve manner) is very computationally inefficient.

This is one of the first indications that simulating a quantum computer on a classical computer might be very difficult.

$$|\psi\rangle = \sum_{x=0}^{2^n-1} a_x |x\rangle$$

$a_x$ are $2^n$ complex numbers

properly normalized: $\sum_{x=0}^{2^n-1} |a_x|^2 = 1$

# n Qubit States

Example:

$$|\psi\rangle = \sum_{x=0}^{2^n-1} a_x |x\rangle \quad a_x = \frac{1}{\sqrt{2^n}}$$

properly normalized: $\sum_{x=0}^{2^n-1} |a_x|^2 = \sum_{x=0}^{2^n-1} \frac{1}{2^n} = 1$

$n = 3$

$$|\psi\rangle = \sum_{x=0}^{7} \frac{1}{2\sqrt{2}} |x\rangle$$

$$= \frac{1}{2\sqrt{2}}(|0\rangle + |1\rangle + |2\rangle + |3\rangle + |4\rangle + |5\rangle + |6\rangle + |7\rangle)$$

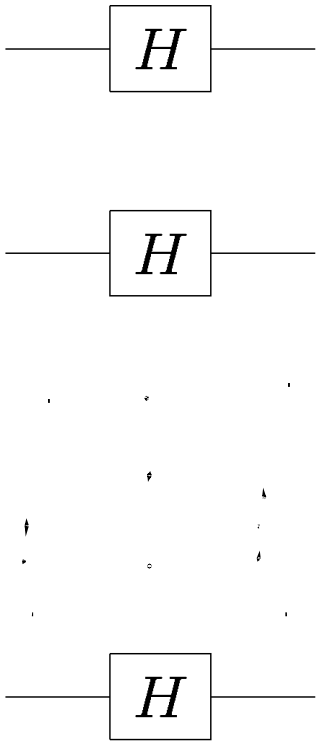
$$= \frac{1}{2\sqrt{2}}(|000\rangle + |001\rangle + |010\rangle + |011\rangle + |100\rangle + |101\rangle + |110\rangle + |111\rangle)$$

Notice how compact this 1st notation

# n Qubit Hadamard

Hadamard all n qubits $H^{\otimes n} = H \otimes H \otimes \cdots \otimes H$

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$




n qubits input

n qubits output

# n Qubit Hadamard

Hadamard one qubit in computational basis:

$$H|x_i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + (-1)^{x_i}|1\rangle) = \frac{1}{\sqrt{2}} \sum_{y_i=0,1} (-1)^{x_i y_i}|y_i\rangle$$

Hadamard n qubits in computational basis:

$$H^{\otimes n}|x\rangle = (H \otimes H \otimes \cdots \otimes H)|x_1\rangle \otimes |x_2\rangle \otimes \cdots \otimes |x_n\rangle$$

$$= (H|x_1\rangle) \otimes (H|x_2\rangle) \otimes \cdots \otimes (H|x_n\rangle)$$

$$= \frac{1}{\sqrt{2}}(|0\rangle + (-1)^{x_1}|1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + (-1)^{x_2}|1\rangle) \otimes \cdots \otimes \frac{1}{\sqrt{2}}(|0\rangle + (-1)^{x_n}|1\rangle)$$

$$= \frac{1}{\sqrt{2^n}} \sum_{y_1,y_2,\ldots,y_n=0}^{1} (-1)^{y_1x_1 + y_2x_2 + \cdots y_nx_n}|y_1, y_2, \ldots, y_n\rangle$$

$$y \cdot x = y_1x_1 + y_2x_2 + \cdots + y_nx_n$$

$$= \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{y \cdot x}|y\rangle$$

# n Qubit Hadamard

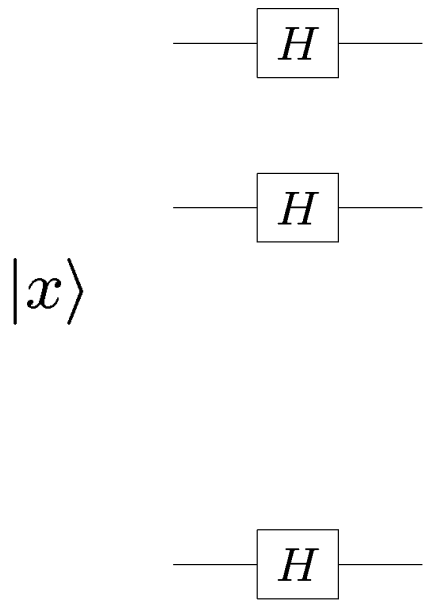
$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{y \cdot x} |y\rangle$$

$$y \cdot x = y_1 x_1 + y_2 x_2 + \cdots + y_n x_n$$

Addition can be done modulo 2 (turns plus to exclusive-or)

$$y \cdot x = y_1 x_1 \oplus y_2 x_2 \oplus \cdots \oplus y_n x_n$$

$$b_1 \oplus b_2 = \begin{cases} 0 \text{ if } b_1 = b_2 \\ 1 \text{ if } b_1 \neq b_2 \end{cases}$$




$$\frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{y \cdot x} |y\rangle$$

Again notice compactness of notation.

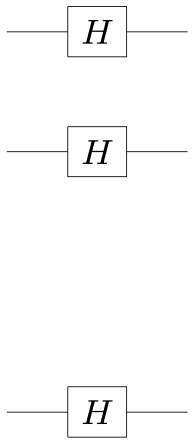
# Superposition Over All

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{y \cdot x} |y\rangle$$

$$y \cdot x = y_1 x_1 \oplus y_2 x_2 \oplus \cdots \oplus y_n x_n$$

If we start in the zero bitstring, then Hadmarding all n qubits creates a superposition over all possible bitstrings:

$$H^{\otimes n}|0\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} |y\rangle$$




$$|0\rangle \qquad \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} |y\rangle$$

# Superposition Over All

Hadamarding the superposition over all states:

$$H^{\otimes n}\left(\frac{1}{\sqrt{2^n}}\sum_{y=0}^{2^n-1}|y\rangle\right) = \frac{1}{\sqrt{2^n}}\sum_{y=0}^{2^n-1}H^{\otimes n}|y\rangle$$

$$= \frac{1}{\sqrt{2^n}}\sum_{y=0}^{2^n-1}\frac{1}{\sqrt{2^n}}\sum_{z=0}^{2^n-1}(-1)^{z\cdot y}|z\rangle$$

$$= \frac{1}{2^n}\sum_{z=0}^{2^n-1}\left(\sum_{y=0}^{2^n-1}(-1)^{z\cdot y}\right)|z\rangle$$

$$\frac{1}{2^n}\sum_{y=0}^{2^n-1}(-1)^{z\cdot y} = \frac{1}{2^n}\sum_{y_1,\ldots,y_n=0}^{1}(-1)^{z_1y_1+\cdots+z_ny_n}$$

# Superposition Over All

$$\frac{1}{2^n}\sum_{y=0}^{2^n-1}(-1)^{z\cdot y} = \frac{1}{2^n}\sum_{y_1,\ldots,y_n=0}^{1}(-1)^{z_1y_1+\cdots+z_ny_n}$$

$$= \frac{1}{2^n}\sum_{y_1,\ldots,y_n=0}^{1}(-1)^{z_1y_1}(-1)^{z_2y_2}\cdots(-1)^{z_ny_n}$$

$$= \left[\frac{1}{2}\sum_{y_1=0}^{1}(-1)^{z_1y_1}\right]\left[\frac{1}{2}\sum_{y_2=0}^{1}(-1)^{z_2y_2}\right]\cdots\left[\frac{1}{2}\sum_{y_n=0}^{1}(-1)^{z_ny_n}\right]$$

$$\frac{1}{2}\sum_{y_i=0}^{1}(-1)^{z_iy_i} = \frac{1}{2}((-1)^0+(-1)^{z_i}) = \delta_{z_i,0}$$

$$\frac{1}{2^n}\sum_{y=0}^{2^n-1}(-1)^{z\cdot y} = \delta_{z_1,0}\delta_{z_2,0}\cdots\delta_{z_n,0} = \delta_{z,0}$$

# Superposition Over All

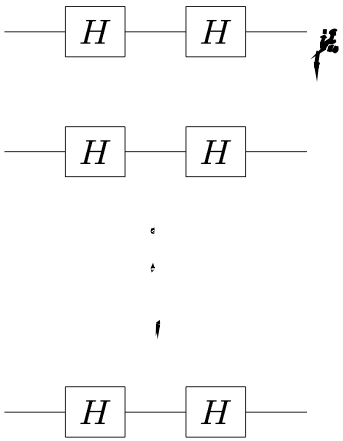
$$H^{\otimes n}\left(\frac{1}{\sqrt{2^n}}\sum_{y=0}^{2^n-1}|y\rangle\right)=\frac{1}{2^n}\sum_{z=0}^{2^n-1}\left(\sum_{y=0}^{2^n-1}(-1)^{z\cdot y}\right)|z\rangle$$

$$\frac{1}{2^n}\sum_{y=0}^{2^n-1}(-1)^{z\cdot y}=\delta_{z_1,0}\delta_{z_2,0}\cdots\delta_{z_n,0}=\delta_{z,0}$$

$$H^{\otimes n}\left(\frac{1}{\sqrt{2^n}}\sum_{y=0}^{2^n-1}|y\rangle\right)=\sum_{z=0}^{2^n-1}\delta_{z,0}|z\rangle=|0\rangle$$

Could have found in easier fashion using

$$(H^{\otimes n})(H^{\otimes n})=(H^2)^{\otimes n}=I$$



# From Comp. Basis to Matrix

From the effect of the Hadamard on the computational basis

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{y \cdot x} |y\rangle$$

We can deduce the form of the matrix in outer product form:

$$H^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x,y=0}^{2^n-1} (-1)^{y \cdot x} |y\rangle\langle x|$$

# Hadamard Basis Elements

Recall that the columns of a matrix form a basis.

What is this basis for the Hadamard?

$$H^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x,y=0}^{2^n-1} (-1)^{y \cdot x} |y\rangle\langle x|$$

The basis elements for the Hadmard are:

$$|\phi_x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{y \cdot x} |y\rangle$$

# Hadamard Basis Elements

Check orthonormality:

$$|\phi_x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{y\cdot x} |y\rangle$$

$$\langle \phi_{x'} | \phi_x \rangle = \frac{1}{\sqrt{2^n}} \sum_{y'=0}^{2^n-1} (-1)^{x'\cdot y'} \langle y' | \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{y\cdot x} |y\rangle$$

$$\langle \phi_{x'} | \phi_x \rangle = \frac{1}{2^n} \sum_{y,y'=0}^{2^n-1} (-1)^{x'\cdot y' + x\cdot y} \langle y | y' \rangle \longleftarrow \delta_{y,y'}$$

$$\langle \phi_{x'} | \phi_x \rangle = \frac{1}{2^n} \sum_{y=0}^{2^n-1} (-1)^{x'\cdot y + x\cdot y}$$

# Hadamard Basis Elements

$$\langle \phi_{x'} | \phi_x \rangle = \frac{1}{2^n} \sum_{y=0}^{2^n-1} (-1)^{x' \cdot y + x \cdot y}$$

$$x' \cdot y + x \cdot y = y_1 x'_1 + y_2 x'_2 + \cdots + y_n x'_n + y_1 x_1 + y_2 x_2 + \cdots + y_n x_n$$

$$= y_1 (x'_1 + x_1) + y_2 (x'_2 + x_2) + \cdots + y_n (x'_n + x_n)$$

$$\langle \phi_{x'} | \phi_x \rangle = \frac{1}{2^n} \sum_{y=0}^{2^n-1} (-1)^{(x' \oplus x) \cdot y}$$

$$x \oplus x' = (x_1 \oplus x'_1, x_2 \oplus x'_2, \ldots, x_n \oplus x'_n)$$

$$\frac{1}{2^n} \sum_{y=0}^{2^n-1} (-1)^{z \cdot y} = \delta_{z_1,0} \delta_{z_2,0} \cdots \delta_{z_n,0} = \delta_{z,0}$$

$$\langle \phi_{x'} | \phi_x \rangle = \delta_{x \oplus x',0} = \delta_{x,x'}$$